Task 1 Internal Force Flow in a Beam: Qualitative

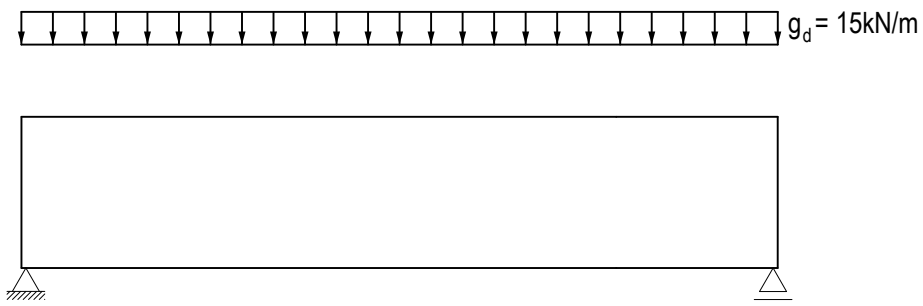
Five beams with the same loading case but with different support conditions are given. Sketch a possible internal force flow for the beams a) to e).

Tip: For b), c) and d), divide the line load at the supports.

Task 2 Internal Force Flow in a Beam: Quantitative

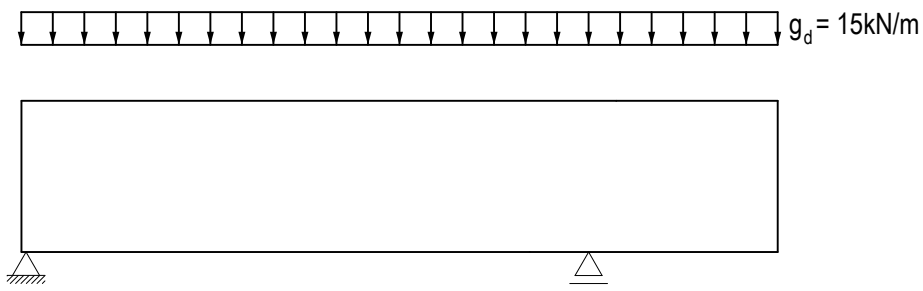
Find the resultant(s) for each of the five beams and determine the global equilibrium with the help of the force diagram. Starting at the largest span width, draw the force diagrams to their internal force flow and construct them neatly in the form diagram. Indicate tension forces with red, compression forces with blue and reaction forces with green.

a)



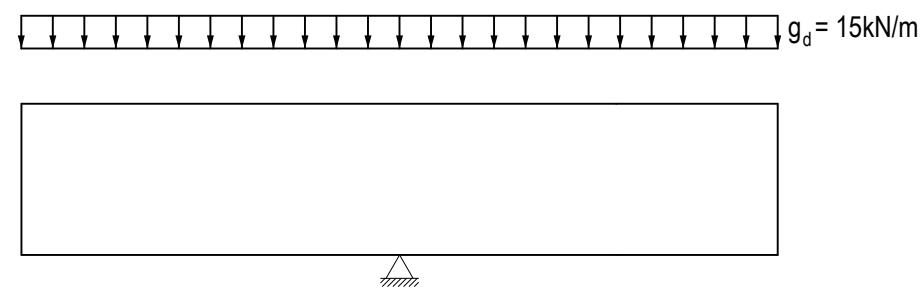
form diagram 1:100 *force diagram 1cm ≙ 20kN*

b)



form diagram 1:100 *force diagram 1cm ≙ 20kN*

c)



form diagram 1:100 *force diagram 1cm ≙ 20kN*